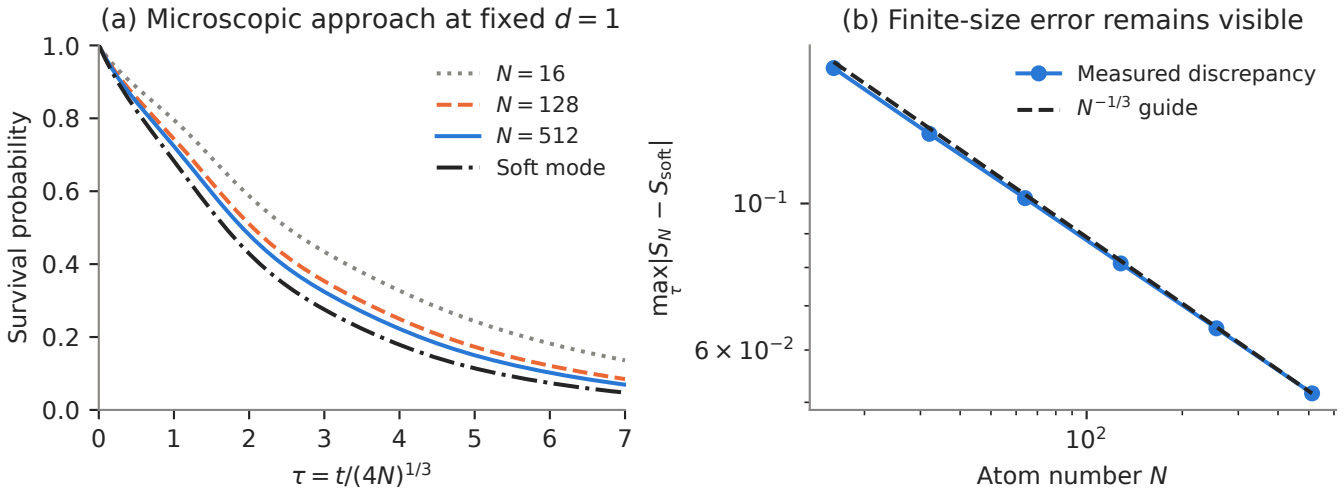
(a) Microscopic approach at fixed $d = 1$
Survival probability
$\tau = t/(4N)^{1/3}$
$N = 16$
$N = 128$
$N = 512$
Soft mode
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5
6
7
(b) Finite-size error remains visible
$\max_\tau |S_N - S_{\mathrm{soft}}|$
Atom number N
Measured discrepancy
$N^{-1/3}$ guide
10^{-1}
6×10^{-2}
10^2